EXPERIMENT 7

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Question 1:

Consider a relation R having attributes R(ABCD), functional dependencies: AB → C, C → D, D → A. Identify the set of candidate keys possible in relation R. List all the set of prime and non-prime attributes.

Answer:
Closures:

- $B^+ = B$
- $AB^+ = ABCD$
- $BC^+ = BCDA$
- $BD^+ = BDAC$

Candidate Keys = **{AB, BC, BD}**
Prime Attributes = **{A, B, C, D}**
Non-Prime Attributes = **{}**

Since all RHS attributes are prime, the relation is in **3NF**.

For BCNF:

- C → D, but C is not a superkey.
- Therefore, BCNF is violated.

Highest Normal Form = 3NF

Question 2:

Relation R(ABCDE) has functional dependencies: A → D, B → A, BC → D, AC → BE. Identify the set of candidate keys possible in relation R. Find the highest normal form.

Answer:
Closures:

- $C^+ = C$
- $AC^+ = ACDBE$
- $BC^+ = BCAD\ E$
- $CD^+ = CD$
- $CE^+ = CE$

Candidate Keys = **{AC, BC}**
Prime Attributes = **{A, B, C}**
Non-Prime Attributes = **{D, E}**

Since $A \rightarrow D$ and A is a proper subset of candidate key AC, there is a **partial dependency** on a non-prime attribute.

Therefore, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 3:

Consider a relation R(ABCDE), with functional dependencies: $B \rightarrow A$, $A \rightarrow C$, $BC \rightarrow D$, $AC \rightarrow BE$. Identify the set of candidate keys possible in relation R. Find the highest normal form.

Answer:
Closures:

- $B^+ = BACDE$
- $A^+ = ACBED$
- $C^+ = C$
- $D^+ = D$
- $E^+ = E$

Candidate Keys = **{A, B}**
Prime Attributes = **{A, B}**
Non-Prime Attributes = **{C, D, E}**

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For BCNF:

- A and B are candidate keys, hence superkeys.
- All non-trivial FDs have a superkey on the left side.

Therefore, the relation satisfies **BCNF**.

Highest Normal Form = BCNF

Question 4:

Consider a relation R(ABCDEF), with functional dependencies: A → BCD, BC → DE, B → D, D → A. Find the highest normal form.

Answer:
Since F never appears on the RHS of any FD, F must be present in every candidate key.

Closures:

- AF^+ = ABCDEF
- BF^+ = ABCDEF
- DF^+ = ABCDEF

Candidate Keys = **{AF, BF, DF}**
Prime Attributes = **{A, B, D, F}**
Non-Prime Attributes = **{C, E}**

Consider A → C:

- A is a proper subset of candidate key AF.
- C is a non-prime attribute.
- Therefore, A → C is a **partial dependency**.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 5:

Relation R(ABCDE) has dependencies: CE → D, D → B, C → A. Find the highest normal form and candidate key set.

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Answer:
Closure:

- CE^+ = CEDBA

Candidate Key = **{CE}**
Prime Attributes = **{C, E}**
Non-Prime Attributes = **{A, B, D}**

Consider C → A:

- C is a proper subset of candidate key CE.
- A is a non-prime attribute.
- Therefore, C → A is a **partial dependency**.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 6:

Relation R(ABCDEF) has dependencies: AB → C, DC → AE, E → F. Find the highest normal form and candidate key set.

Answer:
Since B and D never occur on the RHS, they must be part of every candidate key.

Closures:

- BD^+ = BD
- BDA^+ = BDACEF
- BDC^+ = BDCAEF

Candidate Keys = **{ABD, BCD}**
Prime Attributes = **{A, B, C, D}**
Non-Prime Attributes = **{E, F}**

Consider DC → E:

- DC is a proper subset of candidate key BCD.
- E is a non-prime attribute.

- Therefore, DC → E is a **partial dependency**.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 7:

Relation R(ABCDEFGHI) has dependencies: AB → C, BD → EF, AD → GH, A → I. Find the highest normal form and candidate key set.

Answer:
Since A, B, and D never occur on the RHS, they must be part of every candidate key.

Closure:

- ABD^+ = ABCDEFGHI

Candidate Key = **{ABD}**
Prime Attributes = **{A, B, D}**
Non-Prime Attributes = **{C, E, F, G, H, I}**

Consider AB → C:

- AB is a proper subset of candidate key ABD.
- C is a non-prime attribute.
- Therefore, AB → C is a **partial dependency**.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF

Question 8:

Relation R(ABCDE) has dependencies: AB → CD, D → A, BC → DE. Find the highest normal form and candidate key set.

Answer:
Since B does not appear on the RHS, B must be part of every candidate key.

Closures:

- BC^+ = ABCDE

- $AB^+ = ABCDE$
- $BD^+ = ABCDE$

Candidate Keys = **{BC, AB, BD}**
Prime Attributes = **{A, B, C, D}**
Non-Prime Attributes = **{E}**

For 3NF:

- AB → CD: AB is a superkey.
- D → A: D is not a superkey, but A is a prime attribute.
- BC → DE: BC is a superkey.

Therefore, the relation satisfies **3NF**.

For BCNF:

- D → A, but D is not a superkey.
- Hence, BCNF is violated.

Highest Normal Form = 3NF

Question 9:

Relation R(ABCDE) has dependencies: BC → ADE, D → B. Find the highest normal form and candidate key set.

Answer:
Since C does not appear on the RHS, C must be part of every candidate key.

Closures:

- $BC^+ = ABCDE$
- $CD^+ = ABCDE$

Candidate Keys = **{BC, CD}**
Prime Attributes = **{B, C, D}**
Non-Prime Attributes = **{A, E}**

For 3NF:

- BC → ADE: BC is a superkey.

- D → B: D is not a superkey, but B is a prime attribute.

Therefore, the relation satisfies **3NF**.

For BCNF:

- D → B, but D is not a superkey.
- Hence, BCNF is violated.

Highest Normal Form = 3NF

Question 10:

Relation R(ABC) has dependencies: A → B, B → C, C → A. Find the highest normal form and candidate key set.

Answer:
Closures:

- $A^+ = ABC$
- $B^+ = ABC$
- $C^+ = ABC$

Candidate Keys = **{A, B, C}**
Prime Attributes = **{A, B, C}**
Non-Prime Attributes = **{}**

A, B, and C are all candidate keys and therefore superkeys.

All non-trivial functional dependencies have a superkey on the LHS.

Therefore, the relation satisfies **BCNF**.

Highest Normal Form = BCNF

Question 11:

Consider a relation P having fields F1, F2, F3, F4, F5 with functional dependencies: F1 → F3, F2 → F4, (F1,F2) → F5. In terms of normalization, which normal form is this table in? Describe the prime and non-prime attributes.

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Answer:
Since F1 and F2 never occur on the RHS, they must be part of the candidate key.

Closure:

- $F1F2^+ = F1F2F3F4F5$

Candidate Key = **{F1F2}**
Prime Attributes = **{F1, F2}**
Non-Prime Attributes = **{F3, F4, F5}**

Consider F1 → F3:

- F1 is a proper subset of candidate key F1F2.
- F3 is a non-prime attribute.
- Therefore, it is a **partial dependency**.

Similarly, F2 → F4 is also a partial dependency.

Hence, the relation violates **2NF**.

Highest Normal Form = 1NF